CS 217: Practice questions on unsupervised representation learning

1. Given two uncorrelated zero-mean random variables $X_1 \sim \mathcal{N}(0, 100)$ and $X_2 \sim \mathcal{N}(0, 1)$, what are the principal components of (X_1, X_2) ..3 We need to find the covariance matrix of the random vector $X = [X_1, X_2]^T$.

   The covariance matrix is given by:

$$C = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{bmatrix} \tag{1}$$

   Since, X_1 and X_2 are uncorrelated, $\mathrm{Cov}(X_1, X_2) = 0$. Hence, the covariance matrix is given by:

$$C = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix} \tag{2}$$

   The eigen values of the matrix can be calculated by equating the determinant of the following matrix to 0:

$$\begin{vmatrix} 100 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0 \tag{3}$$

   Therefore, the eigen values are $\lambda_1 = 100$ and $\lambda_2 = 1$. The eigen vectors are given by:

   The corresponding eigen vectors are given by $[1, 0]^T$ and $[0, 1]^T$. Clearly one of the eigen values is much larger than the other. Therefore, the principal component of the data is given by projection of the data on the first eigen vector which is X_1.

2. For more questions on PCA solve exercise 20.4 and 20.5 from the textbook.

3. Show an example dataset or distribution $P(\mathbf{x})$ where the sparse auto-encoder trained with $|\mathbf{z}|_1$ penalty on the code leads to zeroing out of some coordinates all together. In contrast training with coordinate-wise KL penalty $\sum_{k=1}^{L} \mathrm{KL}(p, z_k)$ for a small $p = 0.1$ leads to more coordinates being non-zero but for a small fraction p of instances. L is the length of the code $\mathbf{z} = f_e(\mathbf{x})$ and $\mathbf{x} \in \mathbb{R}^d$ is the input. [You can assume $d = L$] Assume $d = L$ Let $P(\mathbf{x})$ be a Gaussian distribution with diagonal covariance matrix where variance of kth coordinate is σ_k^2. Let the encoder just apply a sigmoid on each $\sigma(\alpha_k x_k + \beta_k)$ where α_k, β_k are trained parameters of the encoder. It can be shown that with $|\mathbf{z}|_1$ penalty, the dimensions correspond to small σ_k values will be zero-ed out, whereas with the KL divergence the model with use up all the dimensions with varying number of non-zeros depending on the data distribution.

4. In the CLIP model for learning to jointly embed an image and text, the loss function for learning image embedding vector I_j of image j, and text embedding vector T_i for text i was designed as:

$$L = -\sum_{i=1}^{B} \log \frac{\exp(I_i.T_i)}{\sum_j \exp(I_j.T_i)} - \sum_{j=1}^{B} \log \frac{\exp(I_j.T_j)}{\sum_r \exp(I_j.T_r)} \tag{4}$$

   Let B denote the batch size.

   (a) Write the gradient of the above loss with respect to the image embedding vector I_1. ..2

(b) Consider a dataset where in the training dataset we have a piece of text T_i co-located with a set of images S_i, such that T_i is applicable to exactly one image in S_i. For such a supervision, how will you train the model? ..3 Now we need to define the score between a set T_i and a set S_i of images, one of which is the correct match. A simple trick is to define this score as $\max_{I \in S_i} I.T_i$. The rest stays the same. We just replaced direct dot-product with max across the elements in the set.